

Reusable architecture knowledge

Prior projects + literature

design artifacts + documents
tool traces + measurements
failures + reviewed results

Current project record

Versioned study state

workload + software versions
constraints + candidate identity
tool, library + process versions
current measurements + failures

Architecture role

supplies intent + constraints
reviews result and forms a
supported recommendation

Commitment authority

decides advance, hold, or reject

AI-native architecture design system



Architecture foundation model

optional learned component
adapted to the current task

Architecture tasks

1. Workload analysis and classification

2. Surrogate performance prediction

3. Candidate software and RTL generation

4. Diagnostic classification of failure



Current project record

specifications + RTL + simulators
+ physical design scripts



Design tools and checks

compilers + simulators + EDA engines
+ formal property checkers
+ independent verification

Empirical evidence

tool returns + traces + DRC reports
+ formal proofs + failed runs
+ rejected alternatives
updates live project state



Supported architecture result

comparison + measurements + assumptions + checks + limits

The moonshot is complete-system capability, not autonomous prompt-to-chip generation.